# Ahmad Faraz Sheikh

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## Experience

**Redbuffer** **October 2024 – Present**
*Machine Learning Engineer* *Islamabad, Pakistan*

**Machine Learning Engineer (L4)** **August 2026 – Present**

**BeyondOne - Market Intelligence Platform**

- Designed and built a Python/FastAPI data ingestion pipeline that continuously fetches articles, news, and macroeconomic metrics from 49 article sources and 6 metric providers across Chile, Saudi Arabia, and global markets using RSS, App Store, SEC EDGAR, and custom scrapers.
- Developed the PostgreSQL schema and persistence layer for 6 normalized tables, implementing SHA-256 content-hash deduplication, deterministic UUID5 identity keys, and upsert-based metric revision tracking.
- Built an asyncio-based per-source scheduler with configurable polling intervals, per-source overlap locking, YAML-driven source matching against a DB-first source registry, and manual trigger endpoints via FastAPI.
- Implemented metric observation revision detection with prior-value tracking and a multi-provider normalization layer that standardizes heterogeneous API responses (REST JSON, SDMX, RSS/Atom XML) into a unified observation model.

**Zonlex RAG Pipeline**

- Developed an end-to-end legal RAG backend using FastAPI, Qdrant, and LiteLLM that ingests municipal code documents, applies structure-aware chunking preserving section IDs and citation anchors, and delivers citation-grounded zoning-code answers with answerability handling.
- Integrated LiteLLM for provider-agnostic generation and embeddings, enabling support for Gemini, vLLM, and other OpenAI-compatible local models through configuration.
- Built an evaluation module using gold Q&A spreadsheets to measure citation recall, answerability accuracy, and answer correctness with LLM-as-judge scoring, producing Excel and JSON reports for end-to-end RAG quality analysis.

**Machine Learning Engineer (L3)** **October 2024 – July 2026**

**Agentic AI Platform**

- Implemented a dynamic and scalable LLM-based guardrails pipeline for the agentic chat system, enabling validation at both tool and input/output levels. Developed an API and integrated a web interface allowing non-technical users to easily add and manage custom guardrails.
- Enhanced the agent with context summarization capabilities to maintain conversations when session history grows large.
- Integrated memory management tools that allow the agent to persist and recall user-specific information across sessions upon request.
- Developed agentic-MCP management functionality enabling the agent to autonomously decide when to initialize or reinitialize agent instances based on task context.
- Integrated Data Masker to the agentic chat, allowing users to mask sensitive data before LLM processing to ensure privacy and data protection.

**AI Hub for Worksite Safety Management**

- Developed and deployed Retrieval-Augmented Generation (RAG) systems by building knowledge bases and integrating them with LLM pipelines.
- Worked with AWS services (Lambda, API Gateway, OpenSearch, S3, ECS, ECR, Terraform) to deploy machine learning pipelines and APIs.
- Designed and implemented scalable data ingestion pipelines for specific domain datasets, including handling de-duplication, indexing, and OpenSearch mapping.
- Designed and developed a near real-time computer vision solution focused on detecting risks in worksite videos and suggesting safety measures to mitigate them.

**Continuas AI - Machine Learning Engineer**

- Designed and implemented an LLM-based reminder generation pipeline that classified thread state, extracted scheduling intent, and generated follow-up reminders with timezone and weekend-aware logic.
- Built a multi-step state machine for AI inference orchestration, including classification, tone analysis, delay calculation, cold-email follow-up handling, and fallback logic for failure recovery.

- Created AI-generated reply and reply-all template generation for email threads, including tone-aware variants tailored to user preferences and recipient context.
- Built automated email tagging and priority classification flows to label conversations and improve thread organization.
- Added structured error handling, observability using Langfuse, and Sentry-based monitoring for model/API failures, timeout handling, and production debugging.

**ML-Driven CAD and Survey Point Reconciliation Platform**

- Developed a GeoGNN/GraphSAGE pipeline with CLIP and all-MiniLM semantic embeddings for matching raw survey points to CAD/DWG geometry using spatial, semantic, and field-code features.
- Built Python scripts for survey-to-CAD comparison, point classification, field-code matching, movement detection, and deduplication.
- Implemented grid search workflows with fault-tolerant resume logic and hyperparameter tuning, achieving validation F1 scores of around 0.69.

**EDI(Electronic Data Interchange) Automation Tool**

- Built an Electronic Data Interchange automation platform designed to validate and generate field-level mappings between input and output documents across multiple trading partners.
- Evaluated and integrated large-context open-source LLMs to enhance mapping accuracy and scalability.
- Deployed the complete platform on-premises using Docker containers and integrated open-source LLMs through vLLM for optimized inference and scalability.

**PanaceaLogics** **September 2024 – October 2024**

*Machine Learning Intern* *Islamabad, Pakistan*

- Conducted research on Machine Learning (ML) and Deep Learning (DL) models to explore their applications and how they can be useful for our task.
- Worked with datasets to train, test, and evaluate ML models for object detection, length, and distance estimation of that object.

## Technical Skills

**Languages / Technologies**: Python, C/C++, SQL, LLM's, RAG's, Docker, Terraform, MCP, AgenticAI
**Deployment Tools**: vLLM, AWS (Lambda Function, API Gateway, Opensearch, ECS, ECR)
**Libraries/Frameworks**: OpenAI Agent SDK, LangGraph, LangChain, Langfuse, Sentry, Numpy, Pandas, Matplotlib, Seaborn, Scipy, Scikit-Learn, NLTK, Tensorflow, Keras, OpenCV, PyTorch

## Education

**Pakistan Institute of Engineering and Applied Sciences** **Sep. 2020 – June 2024**

*Bachelor of Science in Computer Science* *Islamabad, ICT*

- **CGPA:** 3.53/4.00
- **Relevant Coursework:**

  Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision, Data Mining, Data Structures, Object Oriented Programming and Design, Software Engineering, Operating System, Database Systems,

## University Projects

**Human Pose Estimation using DNN (Final Year Project)**

- Explored and implemented various pose estimation techniques, applying Vision Transformer (ViT) on the COCO dataset to achieve enhanced accuracy.
- Adapted ViT for low-light datasets like ExlPose, leading to performance improvements in challenging conditions.
- Developed a robust API for seamless integration of the pose estimation model with mobile applications.
- Built a mobile app using Flutter, integrating the advanced pose estimation model through the custom API for efficient performance.

**Computer Vision Based Geolocation** | *github.com/FarazSheikh16/cv-based-geolocation*

- Trained and fine-tuned MobileNet in Keras using a diverse dataset of University of Salford buildings, leveraging features such as color, texture, and shape to develop an accurate model achieving 94% accuracy for geolocating images. Integrated the model with Google Maps as well

**DNN Based Bot-Account Detection** | *github.com/FarazSheikh16/DNNforBotDetection*

- Implemented two types of Bot detection i.e Account level and Tweet Level.
- Used DNN with SMOTE technique for Account-Level Classification leading to high accuracy.
- For Tweet-Level Classification used LSTM and DNN it also leads to good performance metrics i.e. 80% accuracy.

**Urdu HateSpeech Detection** | *github.com/FarazSheikh16/Urdu-Hate-Speech-Detection*

- Tackled the challenge of spotting offensive content in Urdu, making online spaces safer using ML and DL.
- Achieved over 80% accuracy through rigorous testing, proving the model's effectiveness.
- Improved the models by adding more data, including comments from YouTube and feedback from E-shopping sites, making it better at identifying hate speech in real-world scenarios.

## Leadership and Extracurricular

**Machine Learning Lead** — **Google Developer Student Club, PIEAS**
*September 2023 – June 2024*

**Treasurer** — **PIEAS Artificial Intelligence Society**
*December 2022 – May 2024*

**Marketing Lead** — **Google Developer Student Club, PIEAS**
*August 2022 – June 2023*

**Information Secretary** — **Islamic Society, PIEAS**
*April 2022 – June 2023*